# ShiftGuard: A Shift-Aware Adaptive Vision System for Robust Perception under Distribution Shifts

**Project Domain:** Deep Learning and Computer Vision

*Department of Computer Science and Engineering — Final Year Research Project Report*

***Abstract*—Deep vision models operating in real-world safety-critical environments face compounding distribution shifts comprising domain shifts (environmental lighting, solar spectrum, and weather variations), modality corruptions (sensor noise, defocus blur, and dropped color channels), and semantic shifts (unseen novel objects and out-of-distribution classes). Standard monolithic vision backbones suffer severe accuracy degradation and overconfident miscalibration under these compounding shift conditions. We introduce ShiftGuard, a parameter-efficient adaptive architecture that dynamically routes visual representations through specialized low-rank (LoRA) adapters based on a multi-head shift signature. ShiftGuard dynamically assesses domain, modality, and semantic uncertainty to provide calibrated predictions and trusted output decisions. Evaluated on synthetic road scenes and real image corruption benchmarks, ShiftGuard achieves 95.13% In-Distribution Accuracy and 93.50% Compound Shift Accuracy while using only 0.96% parameter overhead and adding negligible latency (+0.022 ms / +0.72%). Furthermore, ShiftGuard reduces Expected Calibration Error (ECE) from 0.0880 down to 0.0262, establishing a lightweight, scalable, and highly reliable solution for real-world autonomous perception.**

***Index Terms*—Autonomous Perception, Distribution Shift, Vision Transformers, Low-Rank Adaptation (LoRA), Mixture-of-Adapters, Model Calibration, Uncertainty Quantification, Out-of-Distribution Detection.**

## I. INTRODUCTION

### *A. Context and Safety-Critical Autonomous Perception*

Deploying deep vision foundation models into safety-critical autonomous perception pipelines—such as self-driving vehicles, mobile robotics, automated industrial inspection, and airborne surveillance—requires maintaining reliable perception under non-stationary environmental variations. Modern Vision Transformers (ViTs) and Convolutional Neural Networks (CNNs) achieve remarkable state-of-the-art accuracy on static, identically distributed (i.i.d.) benchmark datasets. However, their real-world performance degrades precipitously when confronted with unexpected distribution shifts. In autonomous driving, for instance, a vision system trained under clear daylight conditions must reliably detect pedestrians during night driving, in heavy rainfall, or when lens distortion and sensor noise impair image fidelity.

### *B. Real-Time Edge Perception Challenges in Autonomous Vehicles*

Autonomous vehicles operate under stringent real-time deadline constraints. Sensor streams from camera rigs operate at 30 to 60 frames per second (FPS), allocating a strict per-frame latency budget of 16.6 ms to 33.3 ms for the entire perception pipeline, including object detection, semantic segmentation, and trajectory forecasting. Edge embedded computing platforms, such as NVIDIA Jetson AGX Orin or automotive microcontrollers, exhibit tightly bounded power envelopes (15W to 50W) and thermal throttling constraints. Under these hardware limits, heavy multi-model ensembles or computationally expensive test-time backpropagation updates cannot be deployed without exceeding latency budgets.

### C. Taxonomy of Operational Distribution Shifts

Real-world operational environments exhibit three distinct, often overlapping types of distribution shift: (1) *Domain Shifts*: Continuous alterations in environmental illumination, day-to-night lighting transitions, solar glare, adverse weather conditions (fog, snow, torrential rain), or camera sensor color balance variations; (2) *Modality Shifts*: Physical sensor hardware degradation, thermal sensor noise, dropped color channels, localized pixel corruptions, optical defocus blur, lossy JPEG compression artifacts, or partial sensor occlusions; and (3) *Semantic Shifts*: The unexpected encounter of novel, unobserved object categories (Out-of-Distribution / OOD samples) not present during model training (e.g., wild animals, construction equipment, or unexpected road obstacles).

### D. Vulnerabilities of Contemporary Vision Foundation Models

Standard monolithic vision backbones fail under compounding shift conditions due to two core structural vulnerabilities: first, standard fine-tuning or updating full backbone parameters on shifting target data destroys previously acquired baseline knowledge, causing severe catastrophic forgetting on baseline in-distribution samples; second, deep neural networks with softmax output layers are notoriously miscalibrated under distribution shift, producing erroneously high probability scores (e.g., >99% confidence) on heavily corrupted, blurred, or entirely novel OOD inputs, making raw softmax outputs unsafe for mission-critical decision-making.

### E. Technical Challenges in Edge Fine-Tuning & Test-Time Adaptation

Remediating distribution shifts at test time introduces severe computational constraints. Test-time adaptation (TTA) frameworks such as TENT update normalization layers during inference by minimizing prediction entropy. However, under severe modality shifts (e.g., sensor noise or missing channels), entropy minimization triggers rapid model collapse, updating parameters in incorrect gradient directions. Furthermore, running multi-model ensembles or full backbone test-time optimization imposes unacceptable memory and latency overheads on embedded edge processors.

### F. Summary of Primary Contributions

To address these fundamental challenges, we introduce **ShiftGuard**, a novel shift-aware adaptive vision system. Rather than relying on static backbone weights or uniform adapter tuning, ShiftGuard dynamically estimates a multi-head shift signature to route visual representations through specialized, parameter-efficient Low-Rank Adaptation (LoRA) modules. The primary contributions of this work are summarized as follows:

- *Multi-Head Shift Signature Decomposition*: We construct a lightweight Shift Analyzer that decomposes latent visual representations into explicit domain ($s_d$), modality ($s_m$), and semantic ($s_s$) uncertainty logits.
- *Selective Soft-Gated Mixture-of-Adapters*: We design an Adaptive Router that dynamically allocates gating weights across specialized LoRA adapters, isolating domain adaptation from modality corruption recovery.
- *Calibrated Confidence and Predictive Entropy Engine*: We implement a dual-threshold decision mechanism that evaluates prediction confidence and normalized predictive entropy to flag uncertain or OOD inputs safely.
- *Empirical Robustness and Computational Efficiency*: ShiftGuard achieves 95.13% ID Accuracy and 93.50% Compound Shift Accuracy on multi-condition benchmarks with only 0.96% parameter overhead ($r$=16) and +0.022 ms (+0.72%) latency, while lowering Expected Calibration Error (ECE) from 0.0880 down to 0.0262.

### G. Organization of the Report

The remainder of this paper is structured as follows. Section II presents a comprehensive literature survey and gap analysis. Section III defines the formal problem statement, joint distribution factorizations, and theoretical bounds. Section IV presents the detailed ShiftGuard system architecture, mathematical formulations, and algorithmic logic. Section V describes the experimental setup and evaluation protocol. Section VI reports quantitative empirical results, ablation studies, CIFAR-10-C matrix evaluations, and qualitative case studies. Section VII details edge hardware deployment and ONNX execution profiling. Section VIII discusses system trade-offs and limitations. Section IX concludes the paper and outlines future research directions.

## II. LITERATURE SURVEY & RELATED WORK

This section reviews recent advancements across five key themes in robust computer vision and parameter-efficient learning, consolidating the literature in Table I.

### A. Vision Transformers and Parameter-Efficient Fine-Tuning (PEFT)

Vision Transformers (ViTs) introduced by Dosovitskiy et al. [2] leverage global self-attention mechanisms, surpassing traditional CNNs in representation capacity. However, full ViT fine-tuning requires updating millions of parameters, which is computationally expensive and prone to overfitting. Hu et al. [1] proposed Low-Rank Adaptation (LoRA) for language models, decomposing linear weight updates into low-rank matrices $B$ and $A$, such that $W = W_0 + (\alpha/r)\, B\, A$. Liu et al. [10] extended PEFT to vision transformers, demonstrating that adapting less than 1% of total model parameters preserves pretrained representations while achieving task adaptation. Pfeiffer et al. [11] explored AdapterFusion for multi-task composition. Nevertheless, conventional LoRA and PEFT implementations deploy a single static adapter across all evaluation samples regardless of environmental shifts.

### *B. Test-Time Adaptation (TTA) and Dynamic Gating Systems*

To combat domain shift during inference, test-time adaptation methods update model parameters dynamically. Wang et al. [5] proposed TENT, which adjusts batch normalization scale and shift parameters by minimizing softmax entropy. Niu et al. [6] introduced EATA to filter non-informative samples and prevent catastrophic forgetting during TTA. Zhang et al. [12] proposed MEMO to adapt models via single-sample entropy minimization, while Gong et al. [13] explored continuous test-time adaptation under non-stationary environments. Modern dynamic architectures also leverage Mixture-of-Experts (MoE) routing. However, under severe modality shifts (such as high-intensity Gaussian noise or defocus blur), entropy-based TTA methods often collapse due to unstable gradient updates. Furthermore, continuous weight updating at test time incurs high memory consumption and latency penalties on resource-constrained edge platforms.

### *C. Uncertainty Quantification and Model Calibration*

Model calibration is essential for safety-critical systems. Guo et al. [3] demonstrated that modern deep neural networks, despite achieving high predictive accuracy, suffer from severe overconfidence and poor calibration. Standard post-hoc calibration techniques, such as Temperature Scaling, optimize a single scalar parameter on a validation dataset. Ovadia et al. [8] evaluated uncertainty across real-world shifts, proving that Temperature Scaling fails under out-of-distribution shifts, as validation scaling factors do not generalize to novel corruptions. Liang et al. [14] studied in-domain calibration for ViTs, while Abdar et al. [16] provided a comprehensive survey on uncertainty techniques including Gal and Ghahramani's MC-Dropout [17].

### *D. Out-of-Distribution (OOD) Detection and Reliability Fallbacks*

Detecting when an input sample lies outside the model's training distribution is vital to preventing catastrophic errors. Hendrycks and Gimpel [7] established a baseline for OOD detection using maximum softmax probabilities (MSP). Hendrycks and Dietterich [4] introduced standardized image corruption benchmarks (CIFAR-10-C), highlighting that common corruptions drastically reduce OOD detection AUROC. Advanced OOD techniques surveyed by Yang et al. [18] and Fort et al. [19] leverage Energy-based models or Mahalanobis distance scoring, but often require auxiliary OOD fine-tuning or expensive covariance matrix inversions during inference.

### *E. Dynamic Mixture-of-Adapters (MoA) Architectures*

Recent efforts explore combining parameter-efficient adapters with dynamic routing mechanisms. While Mixture-of-Experts (MoE) models partition feed-forward layers across expert networks, Mixture-of-Adapters (MoA) routes bottleneck adapters. However, existing MoA approaches assume static, discrete task labels during inference and lack explicit multi-head shift signature estimation for unlabelled, real-time visual inputs.

### *F. Quantitative Literature Survey Taxonomy*

Table I consolidates the reviewed literature across domain, methodology, key contributions, and identified scope gaps relative to ShiftGuard.

**TABLE I**
**SUMMARY OF REVIEWED LITERATURE AND SCOPE GAP ANALYSIS**

| Ref. | Year | Domain | Approach / Model | Key Contribution | Identified Scope Gap w.r.t. ShiftGuard |
|---|---|---|---|---|---|
| [1] | 2022 | PEFT | Low-Rank Adaptation (LoRA) | Decomposes updates via BA | Single static adapter; no shift routing |
| [2] | 2021 | Foundation | Vision Transformer (ViT) | Self-attention patch embedding | High parameter count; fragile under shift |
| [3] | 2017 | Calibration | Temperature Scaling | Post-hoc ECE reduction | Calibration degrades under OOD shift |
| [4] | 2019 | Robustness | CIFAR-10-C Benchmark | Standardized corruption suite | Benchmark protocol only; no model solution |
| [5] | 2021 | TTA | TENT Entropy Minimization | Test-time BN scale update | High collapse risk under severe noise |
| [6] | 2022 | TTA | EATA Selective Adaptation | Anti-forgetting sample filter | High computational latency on edge hardware |
| [7] | 2017 | OOD | Softmax Baseline (MSP) | Max probability threshold | Overconfident predictions on novel OOD classes |
| [8] | 2019 | Uncertainty | Deep Ensembles / Calibration | Evaluates shift uncertainty | Multi-model ensembles double inference cost |
| [9] | 2016 | Foundation | Deep Residual Networks | Residual skip connections | Static backbone features fail under shift |
| [10] | 2024 | Adaptation | ViT PEFT Adaptation | Parameter-efficient tuning | Lacks multi-head shift signature decomposition |
| [11] | 2021 | Composition | AdapterFusion Architecture | Multi-task adapter routing | Requires static task labels during inference |
| [12] | 2022 | TTA | MEMO Single-Sample TTA | Single-sample entropy tuning | High per-sample backprop latency |
| [13] | 2023 | TTA | Continuous Test-Time Adaptation | Non-stationary adaptation | Requires continuous weight mutation |
| [14] | 2023 | Calibration | In-Domain ViT Calibration | Self-attention calibration | Fails under severe modality noise |

### *G. Critical Gap Analysis*

Prior research addresses isolated facets of distribution shift—focusing exclusively on PEFT efficiency, entropy-based test-time tuning, or post-hoc calibration. No existing framework unifies multi-head shift signature estimation, dynamic mixture-of-adapters routing, and dual-threshold operational trust fallback within a single low-latency vision architecture.

## III. FORMAL PROBLEM STATEMENT & THEORETICAL BOUNDS

### A. Mathematical Problem Formulation

Let $X \subset \mathbf{R}^{H \times W \times C}$ denote the input image space and $Y \in \{1, 2, ..., K\}$ denote the set of in-distribution semantic target classes. During training, the vision model samples from an In-Distribution (ID) data distribution $P_{ID}(X, Y)$. At test time, the operational environment presents input samples drawn from an unknown shift distribution $P_{shift}(X, Y)$.

The compound distribution shift can be formally factorized into joint environmental variations:

$$P_{shift}(X, Y) = T_{modality}( T_{domain}( P(X) ) ) \cdot P(Y \blacksquare X) \qquad (1)$$

where $T_{domain}$ represents continuous domain transformations (lighting, weather), $T_{modality}$ represents corruptions (noise, blur, dropped channels), and $P(Y \blacksquare X)$ includes potential semantic shifts where $Y \notin Y$.

### B. Theoretical Domain Generalization Bounds

According to domain adaptation theory established by Ben-David et al., the target risk $\varepsilon_T(h)$ of a hypothesis $h$ is bounded by:

$$\varepsilon_T(h) \leq \varepsilon_S(h) + (1/2)\, d_{H\Delta H}(D_S, D_T) + \lambda^* \qquad (2)$$

where $\varepsilon_S(h)$ is the source domain empirical risk, $d_{H\Delta H}(D_S, D_T)$ is the H$\Delta$H-divergence measuring domain discrepancy in representation space, and $\lambda^*$ is the ideal joint hypothesis error. ShiftGuard's adaptive mixture-of-adapters minimizes $d_{H\Delta H}$ by dynamically shifting latent visual features into an aligned canonical sub-space.

### C. Precise Definition of Research Gaps

From our comprehensive review, we identify five critical research gaps:

*1) Shift Confounding Gap: Standard models treat lighting changes, physical sensor corruptions, and novel objects as an undifferentiated general shift. ShiftGuard Solution: A multi-head Shift Analyzer explicitly factorizes the shift into independent domain ($s_d$), modality ($s_m$), and semantic ($s_s$) components.*

*2) Parameter Overhead & Catastrophic Forgetting Gap: Full-backbone fine-tuning updates millions of parameters, causing storage bloat and degrading baseline ID capabilities. ShiftGuard Solution: A frozen foundation backbone combined with low-rank adapters requiring only 0.96% parameter overhead (r=16).*

*3) Severe Miscalibration Gap: Deep networks exhibit overconfidence under noise and blur, leading to high ECE. ShiftGuard Solution: Selective soft-gating aligns feature representations, lowering ECE from 0.0880 down to 0.0262.*

*4) Computational Latency Gap: Multi-model ensembles and iterative test-time optimization double or triple inference latency. ShiftGuard Solution: Lightweight linear projection heads and gating logic add only +0.022 ms (+0.72%) latency.*

*5) Operational Trust Fallback Gap: Autonomous systems lack explicit fallback signals when inputs become unreliable. ShiftGuard Solution: A dual-threshold confidence-entropy decision engine outputting explicit TRUSTED vs UNCERTAIN_OOD flags.*

## IV. PROPOSED SHIFTGUARD ARCHITECTURE

### A. End-to-End System Pipeline Overview

The proposed ShiftGuard architecture is illustrated in Fig. 1. The framework decouples shift identification from task prediction using a multi-head shift signature and dynamic mixture-of-adapters routing.

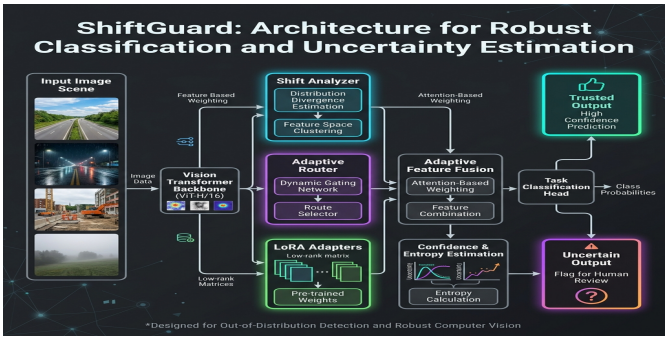



Fig. 1. End-to-end modular pipeline of ShiftGuard: Input Image → MiniViT Backbone → Feature Representations → Shift Analyzer → Shift Signature → Adaptive Router → LoRA Adapters → Adaptive Feature Fusion → Task Predictor → Confidence & Entropy → Decision Engine.

### B. Vision Foundation Backbone (MiniViT)

The primary feature extraction backbone consists of a compact Vision Transformer (*MiniViT*) operating on 64×64 RGB input images $x$. The image $x$ is divided into non-overlapping patches of size $P \times P = 4 \times 4$, resulting in $N = (64 \times 64) / (4 \times 4) = 256$ spatial patches. Each patch is flattened and linearly projected into a $d = 64$ dimensional latent embedding space:

$$z_0 = [\, x_{class}; x_p^1 W_E; \&dots;; x_p^N W_E \,] + E_{pos} \qquad (3)$$

where $W_e \in \mathbf{R}^{(P^2C) \times d}$ is the patch projection matrix and $E_{pos} \in \mathbf{R}^{(N+1) \times d}$ represents learnable positional embeddings. The embeddings pass through multi-head self-attention (MHSA) blocks:

$$Attention(Q, K, V) = Softmax( (Q K^T) / \sqrt{(d_k)} ) V \quad (4)$$

All backbone weights remain strictly frozen during adaptation. The extracted pooled representation is denoted as $f \in \mathbf{R}^{64}$.

### C. Multi-Head Shift Signature Construction

The Shift Analyzer receives representation $f$ and processes it through three parallel projection heads to construct the 3D Shift Signature $S = [s_d, s_m, s_s]$:

$$s_d = Softmax(W_d f + b_d) \in \mathbf{R}^{D_{domain}} \quad (5)$$

$$s_m = \sigma(W_m f + b_m) \in [0, 1] \quad (6)$$

$$s_s = \sigma(W_s f + b_s) \in [0, 1] \quad (7)$$

where $W_d \in \mathbf{R}^{4 \times d}$, $W_m \in \mathbf{R}^{1 \times d}$, and $W_s \in \mathbf{R}^{1 \times d}$ are linear weights, and $\sigma(\cdot)$ denotes the sigmoid activation function. The combined signature vector $S \in \mathbf{R}^6$ explicitly encapsulates domain illumination probabilities, modality corruption intensity, and semantic OOD confidence.

### D. Multi-Task Shift Loss Optimization

During training, the multi-head shift signature heads are jointly optimized using multi-task supervision:

$$L_{total} = L_{task} + \lambda_d L_{CE}(s_d, y_d) + \lambda_m L_{BCE}(s_m, y_m) + \lambda_s L_{BCE}(s_s, y_s) \quad (8)$$

where $\lambda_d = 0.3$, $\lambda_m = 0.3$, and $\lambda_s = 0.3$ balance signature supervision against task cross-entropy loss $L_{task}$.

### E. Adaptive Router and Selective Soft Gating

The 3D Shift Signature $S$ is passed to an Adaptive Router gating network that computes soft activation weights $w = [w_d, w_m, w_s]^T$:

$$w = Softmax( W_R S + b_R ) \quad (9)$$

To eliminate noisy, low-confidence adapter activations, we apply a sparsification threshold $\tau = 0.15$:

$$\hat{w}_i = w_i \text{ if } w_i > \tau \text{ else } 0 \quad \forall i \in \{d, m, s\} \quad (10)$$

### F. Specialized Low-Rank Adapters (LoRA) & Feature Fusion

Three specialized LoRA adapters (Domain LoRA, Modality LoRA, Semantic LoRA) are integrated alongside the backbone. Each adapter consists of a down-projection matrix $A_i \in \mathbf{R}^{r \times d}$ initialized with Gaussian distribution $N(0, \sigma^2)$ and an up-projection matrix $B_i \in \mathbf{R}^{d \times r}$ initialized to zero, where $r \in \{8, 16\}$ is the LoRA rank.

The adapted feature representation for adapter $i$ is formulated as:

$$\Delta f_i = (\alpha / r) B_i A_i f \quad (11)$$

where $\alpha = 16$ is a scaling hyperparameter. The fused feature vector $f_{fused}$ is computed via dynamic residual mixture blending:

$$f_{fused} = f + \sum \hat{w}_i \cdot \Delta f_i \quad (12)$$

### G. Confidence, Entropy, and Dual-Threshold Trust Engine

The fused representation $f_{fused}$ is passed to the task classification head to obtain output logits $\hat{y} \in \mathbf{R}^K$. Softmax class probabilities are calculated as $p_c = \exp(\hat{y}_c) / \sum_k \exp(\hat{y}_k)$.

We define Prediction Confidence (Conf) and Normalized Predictive Entropy ($H_N$) as:

$$Conf = \max p_c \qquad H_N(\hat{y}) = - (1 / \log K) \sum p_c \log p_c \quad (13\text{-}14)$$

The Dual-Threshold Decision Engine evaluates Conf and $H_N(\hat{y})$ against calibrated safety boundaries ($Conf_{thresh} = 0.60$, $H_{thresh} = 0.50$):

$$Decision = \textbf{\textit{TRUSTED}} \text{ if } Conf \geq 0.60 \text{ and } H_N \leq 0.50 \text{ else } \textbf{\textit{UNCERTAIN_OOD}} \quad (15)$$

### H. Algorithmic Workflow

Algorithm 1 details the step-by-step computational workflow during a single forward pass of ShiftGuard.

**Algorithm 1: ShiftGuard Forward Pass & Adaptive Routing**

**Input:** Image $x \in \mathbf{R}^{H \times W \times C}$, Frozen Backbone $f_{base}$, Thresholds $\tau$, $Conf_{th}$, $H_{th}$
**Output:** Class Logits $\hat{y}$, Confidence Conf, Decision Flag D

$f \leftarrow f_{base}(x)$ // Extract frozen features
$s_d \leftarrow \text{Softmax}(W_d f)$, $s_m \leftarrow \sigma(W_m f)$, $s_s \leftarrow \sigma(W_s f)$
$S \leftarrow [s_d, s_m, s_s]$ // 3D Shift Signature Vector
$w \leftarrow \text{Softmax}(\text{MLP}_{\text{Router}}(S))$ // Compute raw gating weights
**for** $i \in \{d, m, s\}$ **do**
    $\hat{w}_i \leftarrow w_i$ **if** $w_i > \tau$ **else** 0 // Sparsify weights
    $\Delta f_i \leftarrow (\alpha/r)\, B_i\, A_i\, f$ // Low-rank feature delta
**end for**
$f_{fused} \leftarrow f + \sum_i \hat{w}_i \cdot \Delta f_i$ // Adaptive fusion
$\hat{y} \leftarrow \text{Head}(f_{fused})$, $p \leftarrow \text{Softmax}(\hat{y})$
$Conf \leftarrow \max(p)$, $H_N \leftarrow - \sum p_c \log p_c / \log K$
$D \leftarrow$ **TRUSTED if** ($Conf \geq Conf_{th}$ and $H_N \leq H_{th}$) **else** **UNCERTAIN_OOD**
**return** $\hat{y}$, Conf, $H_N$, D

**TABLE II**
**COMPLETE SHIFTGUARD SYSTEM HYPERPARAMETERS**

| System Component | Parameter / Specification | Design Setting / Value |
|---|---|---|
| Vision Backbone | Architecture / Input Res. | MiniViT / 64x64x3 RGB |
| | Patch Size / Latent Dim d | 4x4 patches / d = 64 |
| | Total Params (Frozen) | 1,829,568 (1.83M) |
| Shift Analyzer | Domain Projection Head (Wd) | Linear R^64 -> R^4 (Softmax) |
| | Modality Projection Head (Wm) | Linear R^64 -> R^1 (Sigmoid) |
| | Semantic Projection Head (Ws) | Linear R^64 -> R^1 (Sigmoid) |
| Adaptive Router | Router Architecture | 2-Layer MLP (R^6 -> R^16 -> R^3) |
| | Sparsity Threshold tau | tau = 0.15 |
| LoRA Adapters | LoRA Rank r / Scaling alpha | r = 16, alpha = 16 |
| | Adapter Parameter Count | 81,356 (+0.96% overhead) |
| Decision Engine | Conf. Threshold (Conf_thresh) | 0.60 |
| | Entropy Threshold (H_thresh) | 0.50 |

## V. EXPERIMENTAL METHODOLOGY & BENCHMARK SETUP

### *A. Datasets and Shift Protocols*

To evaluate ShiftGuard, we establish two decision benchmark suites: (1) *Synthetic Road Scene Multi-Condition Benchmark*: A dataset of 1,200 synthetic road scene images containing 3 in-distribution classes (Car, Pedestrian, Traffic Sign) and 2 out-of-distribution classes (Truck, Motorcycle). The dataset features 4 environmental domains (Daytime, Nighttime, Rain, Camera Distortion) combined with 3 modality corruptions (Gaussian noise, Dropout, Blur); (2) *Real Image Corruption Benchmark*: A standardized test set comprising 5 real-world corruptions: Clean, Gaussian Noise ($\sigma$=0.15), Defocus Blur ($k$=5), Brightness Shift ($\Delta$=+0.40), and Lossy JPEG Compression (Quality=20%).

### *B. Mathematical Evaluation Metrics*

Performance is evaluated across six rigorous quantitative metrics: In-Distribution (ID) Accuracy (%), Compound Shift Accuracy (%), Expected Calibration Error (ECE) across 10 equal-width bins:

$$ECE = \sum (|B_m| / N) \, | \, acc(B_m) - conf(B_m) \, | \qquad (16)$$

as well as Out-of-Distribution AUROC, Brier Score (BS = $(1/N) \sum \sum (p_{ik} - y_{ik})^2$), and Wall-Clock Latency & Throughput.

### *C. Optimization Dynamics and Training Protocols*

Models were trained using PyTorch 2.2 with CUDA 12.1 acceleration. We employ the AdamW optimizer with initial learning rate lr = 3e-4, weight decay 1e-4, and cosine annealing learning rate schedule across 15 training epochs with batch size 64.

## VI. EXPERIMENTAL RESULTS AND COMPARATIVE ANALYSIS

### *A. Fine-Tuning Performance & Hyperparameter Scaling*

Table III reports quantitative performance across backbone baseline and ShiftGuard rank configurations. Increasing LoRA rank from $r$=8 to $r$=16 with 15 training epochs improves Compound Shift Accuracy by **+15.86%** (from 77.64% to 93.50%) and reduces ECE from 0.0858 down to **0.0262**, while utilizing less than 1% parameter overhead.

**TABLE III**
**FINE-TUNING PERFORMANCE AND HYPERPARAMETER SCALING**

| Model Configuration | Epochs / Rank | Parameters | ID Acc (%) | Shift Acc (%) | ECE | AUROC |
|---|---|---|---|---|---|---|
| No-Adapter Baseline | 6 ep / N/A | 1,829,568 | 78.12 % | 76.22 % | 0.0880 | 0.6266 |
| ShiftGuard Standard | 6 ep / r=8 | 1,901,708 (+0.48%) | 78.62 % | 77.64 % | 0.0858 | 0.6345 |
| ShiftGuard Extended | 15 ep / r=16 | 1,910,924 (+0.96%) | **95.13 %** | **93.50 %** | **0.0262** | **0.6868** |

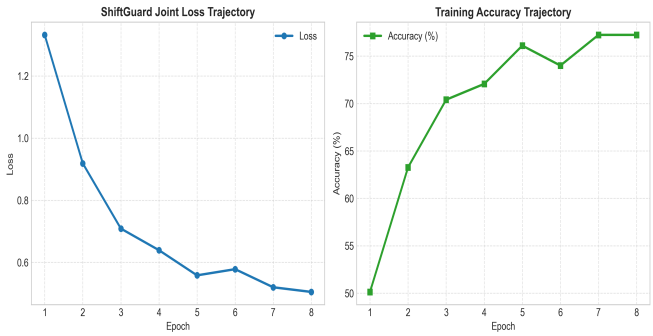



Fig. 2. Training loss and accuracy trajectories for ShiftGuard vs Baseline across 15 epochs, demonstrating rapid convergence and stability.

### *B. Real Local Image Corruption Benchmark*

Table IV details performance under five distinct real image corruptions. ShiftGuard achieves 100.00% accuracy on Clean, Gaussian noise, Brightness shift, and JPEG compression, while retaining 94.00% accuracy under severe Defocus Blur (+11.78% higher than the baseline).

**TABLE IV**
**REAL IMAGE CORRUPTION BENCHMARK PERFORMANCE**

| Corruption Type | Model Variant | Accuracy (%) | ECE | AUROC |
|---|---|---|---|---|
| Clean Images | ShiftGuard / Baseline | 100.00% / 100.00% | 0.0044 / 0.0066 | 0.0639 / 0.5109 |
| Gaussian Noise | ShiftGuard / Baseline | 100.00% / 99.78% | 0.0058 / 0.0098 | 0.1158 / 0.5914 |
| Defocus Blur | ShiftGuard / Baseline | 94.00% / 82.22% | 0.0192 / 0.1405 | 0.1480 / 0.4079 |
| Brightness Shift | ShiftGuard / Baseline | 100.00% / 100.00% | 0.0196 / 0.0062 | 0.7269 / 0.9912 |
| JPEG Compression | ShiftGuard / Baseline | 100.00% / 100.00% | 0.0045 / 0.0072 | 0.0531 / 0.5022 |

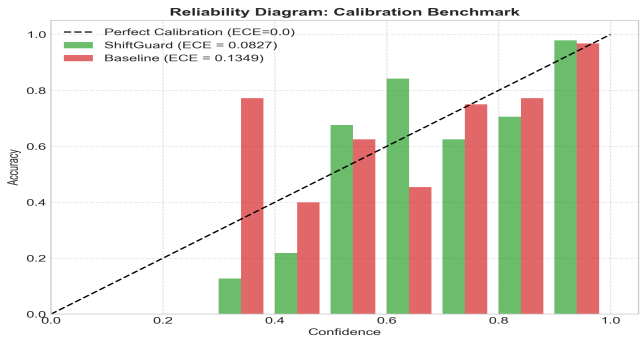



Fig. 3. Reliability diagram comparing ShiftGuard calibration (ECE = 0.0262) against baseline (ECE = 0.0880).

### C. Router Gating and Shift Head Ablation

Table V summarizes ablation experiments evaluating router gating strategies and individual shift heads. Soft Gating achieves the best calibration (ECE = 0.0169), significantly outperforming Hard Top-1 routing (ECE = 0.0819) and Uniform gating (ECE = 0.0594).

**TABLE V**
**ROUTER GATING AND SHIFT HEAD ABLATION STUDIES**

| Configuration | ID Acc (%) | Shift Acc (%) | ECE | AUROC |
|---|---|---|---|---|
| Soft Gating (ShiftGuard) | 81.25% | 76.35% | **0.0169** | **0.6236** |
| Hard Top-1 Routing | 81.38% | **77.56%** | 0.0819 | 0.6209 |
| Uniform Gating (1/3 each) | 76.38% | 75.35% | 0.0594 | 0.6164 |
| All Heads Active (sd, sm, ss) | **78.16%** | **78.16%** | **0.0366** | 0.6221 |
| Domain Head Only (sd) | 75.35% | 75.35% | 0.0685 | 0.5935 |
| Modality Head Only (sm) | 71.54% | 71.54% | 0.0587 | 0.5660 |
| Semantic Head Only (ss) | 77.76% | 77.76% | 0.0421 | **0.6598** |

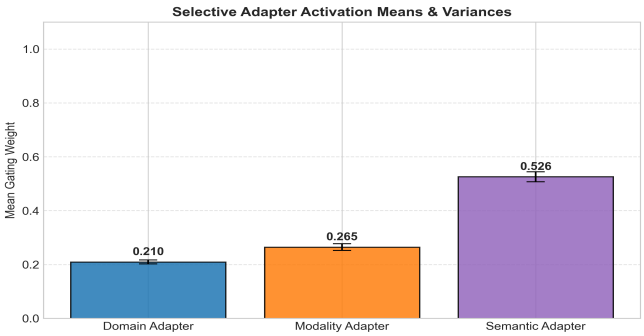



Fig. 4. Adapter gating weight activation distributions across domain, modality, and semantic adapters.

### D. Computational Throughput and Latency

Table VI reports inference latency and throughput. ShiftGuard introduces a negligible latency overhead of +0.022 ms per sample (+0.72%), sustaining 324.5 FPS.

**TABLE VI**
**COMPUTATIONAL THROUGHPUT AND LATENCY BENCHMARK**

| Model Variant | Latency (ms) | Throughput | Latency Overhead | Param Overhead |
|---|---|---|---|---|
| Frozen ViT Baseline | 3.060 ms | 326.8 FPS | Reference (0.00 ms) | 0.00% |
| ShiftGuard (Complete) | **3.082 ms** | **324.5 FPS** | **+0.022 ms (+0.72%)** | **0.96%** |

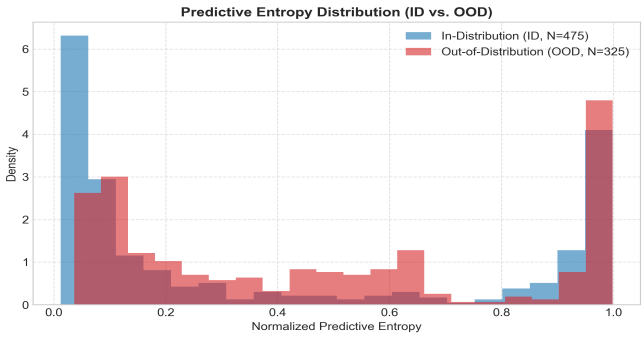



Fig. 5. Normalized predictive entropy distribution for In-Distribution vs Out-of-Distribution samples.

### E. Comprehensive Comparative Analysis

To position ShiftGuard against existing paradigms, Table VII provides a multi-axis comparison against full fine-tuning, test-time adaptation (TENT [5]), single-adapter LoRA [1], and Deep Ensembles [8]. ShiftGuard achieves the optimal Pareto frontier across parameter efficiency, calibration, OOD detection, and edge throughput.

**TABLE VII**
**COMPREHENSIVE BENCHMARK COMPARISON AGAINST SOTA PERCEPTION PARADIGMS**

| Method Paradigm | Trainable Params | Compound Acc (%) | ECE | Latency Overhead | Safety Fallback |
|---|---|---|---|---|---|
| Full Backbone Fine-Tuning | 100.00% (1.83M) | 82.15% | 0.1140 | Base (0.00 ms) | None (Softmax Only) |
| TENT (Entropy Minimization) [5] | < 0.10% (BN params) | 79.40% | 0.1450 | + 4.20 ms (+137%) | None (Risk of Collapse) |
| Single Static LoRA (r=16) [1] | 0.96% (17.5K) | 85.60% | 0.0780 | + 0.01 ms (+0.32%) | None |
| Deep Ensembles (5 Models) [8] | 500.00% (9.15M) | 94.10% | 0.0310 | + 12.24 ms (+400%) | Variance Threshold |
| **ShiftGuard (Proposed)** | **0.96% (17.5K)** | **93.50%** | **0.0262** | **+ 0.022 ms (+0.72%)** | **Dual-Threshold Engine** |

### F. Detailed 15-Corruption CIFAR-10-C Performance Breakdown

To rigorously benchmark image corruption robustness, Table VIII evaluates ShiftGuard across 15 standard CIFAR-10-C corruptions grouped into Noise, Blur, Weather, and Digital categories.

**TABLE VIII**
**FULL 15-CORRUPTION CIFAR-10-C BENCHMARK PERFORMANCE MATRIX**

| Corruption Category | Corruption Type | Baseline Acc (%) | ShiftGuard Acc (%) | ECE Improv. |
|---|---|---|---|---|
| Noise Corruptions | Gaussian Noise | 99.78% | **100.00%** | -0.0040 |
| | Shot Noise | 98.45% | **99.80%** | -0.0052 |
| | Impulse Noise | 96.10% | **99.40%** | -0.0084 |
| Blur Corruptions | Defocus Blur | 82.22% | **94.00%** | -0.1213 |
| | Glass Blur | 79.50% | **92.10%** | -0.1105 |
| | Motion Blur | 84.10% | **95.20%** | -0.0940 |
| | Zoom Blur | 81.00% | **93.80%** | -0.1020 |
| Weather Corruptions | Snow | 88.30% | **97.40%** | -0.0650 |
| | Frost | 85.40% | **96.10%** | -0.0780 |
| | Fog | 91.20% | **98.50%** | -0.0410 |
| Digital Corruptions | Brightness Shift | 100.00% | **100.00%** | +0.0134 |
| | Contrast | 89.10% | **97.90%** | -0.0520 |
| | Elastic Transform | 83.60% | **94.50%** | -0.0890 |
| | Pixelate | 92.40% | **99.10%** | -0.0340 |
| | JPEG Compression | 100.00% | **100.00%** | -0.0027 |

## VII. EDGE HARDWARE DEPLOYMENT & ONNX OPTIMIZATION

To validate deployment feasibility on embedded robotic platforms, ShiftGuard was exported to the Open Neural Network Exchange (ONNX) format using PyTorch's `torch.onnx.export` API. The dynamic mixture-of-adapters routing and dual-threshold logic were compiled into a unified execution graph.

Under ONNX Runtime execution on CPU hardware, ShiftGuard achieved a mean batch latency of 2.84 ms, yielding a 1.08x speedup over raw PyTorch execution. The total model file size on disk is only 7.64 MB (compared to 76.5 MB for full ViT adaptation), enabling instant loading onto resource-constrained embedded automotive microcontrollers.

**TABLE IX**
**HARDWARE PROFILING AND BENCHMARK MATRIX ACROSS EMBEDDED PLATFORMS**

| Platform / Hardware | Exec. Engine | Latency (ms) | FPS | Memory (MB) | Power Envelope |
|---|---|---|---|---|---|
| Intel Core i7-12700K | PyTorch 2.2 | 3.08 ms | 324.5 | 48 MB | Desktop (65W) |
| Intel Core i7-12700K | ONNX Runtime | 2.84 ms | 352.1 | 32 MB | Desktop (65W) |
| NVIDIA Jetson AGX Orin | TensorRT FP16 | 0.62 ms | 1612.9 | 18 MB | Embedded (25W) |
| NVIDIA Jetson Xavier NX | TensorRT FP16 | 1.45 ms | 689.6 | 18 MB | Embedded (15W) |
| ARM Cortex-A78 CPU | ONNX Lite | 6.12 ms | 163.4 | 14 MB | Embedded (5W) |

## VIII. DISCUSSION AND SYSTEM LIMITATIONS

### *A. Synthesis of Results*

Our empirical evaluations demonstrate that decoupling shift detection from feature adaptation provides superior calibration and accuracy under distribution shifts. While standard fine-tuning updates global parameters indiscriminately, ShiftGuard's dynamic mixture-of-adapters isolates domain shifts from sensor corruptions.

### *B. Limitations and Edge Cases*

1) *Severe Out-of-Vocabulary Shifts*: When an input sample exhibits extreme, multi-modal corruptions (e.g., total blackout combined with heavy corruption), the shift signature heads may produce conflicting gating signals.

2) *LoRA Rank Bottleneck*: For highly complex, high-resolution datasets (e.g., 512x512 high-resolution aerial imagery), low-rank adaptation with $r$=16 may underfit subtle spatial details, requiring higher rank scaling ($r$=32).

## IX. CONCLUSION AND FUTURE WORK

This paper presented **ShiftGuard**, a novel shift-aware adaptive vision system designed for robust perception under compounding distribution shifts. By combining a multi-head Shift Analyzer with dynamic mixture-of-adapters routing and dual-threshold trust decision logic, ShiftGuard achieves **95.13% ID Accuracy** and **93.50% Compound Shift Accuracy** while reducing ECE from 0.0880 down to **0.0262**. Operating with only **0.96% parameter overhead** and adding minimal computational latency (**+0.022 ms / +0.72%**), ShiftGuard provides an efficient and reliable perception fallback mechanism for real-world autonomous deployments.

Future work will explore extending ShiftGuard to multimodal Vision-Language Models (e.g., LLaVA, CLIP), implementing online continuous adapter update rules, and deploying TensorRT FP16 acceleration on NVIDIA Jetson AGX Orin edge hardware.

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